

Indexed Files.

Creating an Indexed File

```
$ SET SOURCEFORMAT"FREE"
IDENTIFICATION DIVISION.
PROGRAM-ID.    CreateIndexedFromSeq.
* Creates an indexed file    from a sequential file.

ENVIRONMENT DIVISION.
INPUT-OUTPUT SECTION.
FILE-CONTROL.
    SELECT VideoFile ASSIGN TO "VIDEO.DAT"
        ORGANIZATION IS INDEXED
        ACCESS MODE IS RANDOM
        RECORD KEY IS VideoCode
        ALTERNATE RECORD KEY IS VideoTitle
            WITH DUPLICATES
        FILE STATUS IS VideoStatus.
    SELECT SeqVideoFile ASSIGN TO "INVIDEO.DAT".

DATA DIVISION.
FILE SECTION.
FD VideoFile.
01 VideoRecord.
    02 VideoCode          PIC 9(5).
    02 VideoTitle         PIC X(40).
    02 VideoSupplierCode  PIC 99.
FD SeqVideoFile.
01 SeqVideoRecord.
    88 EndOfFile VALUE HIGH-VALUES.
    02 SeqVideoCode       PIC 9(5).
    02 SeqVideoTitle      PIC X(40).
    02 SeqVideoSupplierCode PIC 99.

WORKING-STORAGE SECTION.
01    VideoStatus          PIC X(2).
```

Creating an Indexed File

```
PROCEDURE DIVISION.
```

```
Begin.
```

```
    OPEN INPUT SeqVideoFile.
```

```
    OPEN OUTPUT VideoFile.
```

```
    READ SeqVideoFile
```

```
        AT END SET EndOfFile TO TRUE
```

```
END-READ.
```

```
    PERFORM UNTIL EndOfFile
```

```
        WRITE VideoRecord FROM SeqVideoRecord
```

```
            INVALID KEY DISPLAY "VIDEO STATUS :- ", VideoStatus
```

```
    END-WRITE
```

```
    READ SeqVideoFile
```

```
        AT END SET EndOfFile TO TRUE
```

```
    END-READ
```

```
END-PERFORM.
```

```
CLOSE VideoFile, SeqVideoFile.
```

```
STOP RUN.
```

Reading an Indexed File - Sequentially.

```
      $ SET SOURCEFORMAT"FREE"
IDENTIFICATION DIVISION.
PROGRAM-ID.    ReadingIndexedFile.
* Sequential reading of an indexed file

ENVIRONMENT DIVISION.
INPUT-OUTPUT SECTION.
FILE-CONTROL.
    SELECT VideoFile ASSIGN TO "VIDEO.DAT"
        ORGANIZATION IS INDEXED
        ACCESS MODE IS DYNAMIC
        RECORD KEY IS VideoCode
        ALTERNATE RECORD KEY IS VideoTitle
            WITH DUPLICATES
        FILE STATUS IS VideoStatus.

DATA DIVISION.
FILE SECTION.
FD VideoFile
01 VideoRecord.
    88 EndOfFile VALUE HIGH-VALUE.
    02 VideoCode          PIC 9(5).
    02 VideoTitle         PIC X(40).
    02 SupplierCode       PIC 99.

WORKING-STORAGE SECTION.
01 VideoStatus            PIC X(2).
01 RequiredSequence      PIC 9.
    88 VideoCodeSequence VALUE 1.
    88 VideoTitleSequence VALUE 2.
01 PrnVideoRecord.
    02 PrnVideoCode       PIC 9(5).
    02 PrnVideoTitle      PIC BBBBX(40).
    02 PrnSupplierCode    PIC BBBB99.
```

Reading an Indexed File - Sequentially.

```
PROCEDURE DIVISION.
```

```
Begin.
```

```
    OPEN INPUT VideoFile.
```

```
    DISPLAY "Enter key : 1=VideoCode, 2=VideoTitle ->"
```

```
        WITH NO ADVANCING.
```

```
    ACCEPT RequiredSequence.
```

```
    IF VideoTitleSequence
```

```
        MOVE SPACES TO VideoTitle
```

```
        START VideoFile KEY IS GREATER THAN VideoTitle
```

```
            INVALID KEY    DISPLAY "VIDEO STATUS :- ", VideoStatus
```

```
        END-START
```

```
    END-IF
```

```
    READ VideoFile NEXT RECORD
```

```
        AT END SET EndOfFile TO TRUE
```

```
    END-READ.
```

```
    PERFORM UNTIL EndOfFile
```

```
        MOVE VideoCode TO PrnVideoCode
```

```
        MOVE VideoTitle TO PrnVideoTitle
```

```
        MOVE SupplierCode TO PrnSupplierCode
```

```
        DISPLAY PrnVideoRecord
```

```
        READ VideoFile NEXT RECORD
```

```
            AT END SET EndOfFile TO TRUE
```

```
        END-READ
```

```
    END-PERFORM.
```

```
    CLOSE VideoFile.
```

```
    STOP RUN.
```

Reading an Indexed File - Sequentially.

```
RUN OF INDEX-EG2.EXE USING VIDEOCODE KEY
Enter key : 1=VideoCode, 2=VideoTitle ->1
00121 FLIGHT OF THE CONDOR, THE 03
00333 PREDATOR 02
00444 LIVING EARTH, THE 03
01001 COMMANDO 02
01100 ROBOCOP 01
02001 LEOPARD HUNTS IN DARKNESS, A 03
02121 DIRTY DANCING 04
03031 COMPETENT CREW 05
03032 YACHT MASTER 05
04041 OPEN OCEAN SAILING 05
04042 PRINCESS BRIDE, THE 06
04444 LIFE ON EARTH 03
05051 OVERBOARD 01
06061 HOPE AND GLORY 07
07071 AMONG THE WILD CHIMPANZEES 03
08081 WHALE NATION 03
09091 BESTSELLER 07
10001 WICKED WALTZING 04
11111 TERMINATOR, THE 02
13301 MASSACRE AT MASAI MARA 03
14032 KNOTTY PROBLEMS FOR SAILORS 05
17001 ALIEN 07
17002 ALIENS 07
17041 GARFIELD TAKES A HIKE 06
18001 SURVIVING THE STORM 05
19444 PINOCCIO 02
```

```
RUN OF INDEX-EG2 USING VIDEOTITLE KEY
Enter key : 1=VideoCode, 2=VideoTitle ->2
17001 ALIEN 07
17002 ALIENS 07
07071 AMONG THE WILD CHIMPANZEES 03
09091 BESTSELLER 07
01001 COMMANDO 02
03031 COMPETENT CREW 05
02121 DIRTY DANCING 04
00121 FLIGHT OF THE CONDOR, THE 03
17041 GARFIELD TAKES A HIKE 06
06061 HOPE AND GLORY 07
14032 KNOTTY PROBLEMS FOR SAILORS 05
02001 LEOPARD HUNTS IN DARKNESS, A 03
04444 LIFE ON EARTH 03
00444 LIVING EARTH, THE 03
13301 MASSACRE AT MASAI MARA 03
04041 OPEN OCEAN SAILING 05
05051 OVERBOARD 01
19444 PINOCCIO 02
00333 PREDATOR 02
04042 PRINCESS BRIDE, THE 06
01100 ROBOCOP 01
18001 SURVIVING THE STORM 05
11111 TERMINATOR, THE 02
08081 WHALE NATION 03
10001 WICKED WALTZING 04
03032 YACHT MASTER 05
```

Reading an Indexed File - Directly.

IDENTIFICATION DIVISION.

PROGRAM-ID. ReadingIndexedFile.

* Illustrates direct read on an indexed file by any key

ENVIRONMENT DIVISION.

INPUT-OUTPUT SECTION.

FILE-CONTROL.

 SELECT VideoFile ASSIGN TO "VIDEO.DAT"

 ORGANIZATION IS INDEXED

 ACCESS MODE IS DYNAMIC

 RECORD KEY IS VideoCode

 ALTERNATE RECORD KEY IS VideoTitle

 WITH DUPLICATES

 FILE STATUS IS VideoStatus.

DATA DIVISION.

FILE SECTION.

FD VideoFile.

01 VideoRecord.

 02 VideoCode PIC 9(5).

 02 VideoTitle PIC X(40).

 02 SupplierCode PIC 99.

WORKING-STORAGE SECTION.

01 VideoStatus PIC X(2).

 88 RecordFound VALUE "00".

01 RequiredKey PIC 9.

 88 VideoCodeKey VALUE 1.

 88 VideoTitleKey VALUE 2.

01 PrnVideoRecord.

 02 PrnVideoCode PIC 9(5).

 02 PrnVideoTitle PIC BBBBX(40).

 02 PrnSupplierCode PIC BBBB99.

Reading an Indexed File - Directly.

```
PROCEDURE DIVISION.  
Begin.  
  OPEN INPUT VideoFile.  
  DISPLAY "Chose key VideoCode = 1,  VideoTitle = 2 ->  "  
    WITH NO ADVANCING.  
  ACCEPT RequiredKey.  
  IF VideoCodeKey  
    DISPLAY "Enter Video Code (5 digits) -> " WITH NO ADVANCING  
    ACCEPT VideoCode  
    READ VideoFile  
      KEY IS VideoCode  
      INVALID KEY  DISPLAY "VIDEO STATUS :- ", VideoStatus  
    END-READ  
  END-IF  
  IF VideoTitleKey  
    DISPLAY "Enter Video Title (40 chars) -> " WITH NO ADVANCING  
    ACCEPT VideoTitle  
    READ VideoFile  
      KEY IS VideoTitle  
      INVALID KEY  DISPLAY "VIDEO STATUS :- ", VideoStatus  
    END-READ  
  END-IF  
  IF RecordFound  
    MOVE VideoCode TO PrnVideoCode  
    MOVE VideoTitle TO PrnVideoTitle  
    MOVE SupplierCode TO PrnSupplierCode  
    DISPLAY PrnVideoRecord  
  END-IF.  
  CLOSE VideoFile.  
  STOP RUN.
```


Reading an Indexed File - Directly.

```
RUN OF INDEX-EG3.EXE USING VIDEOCODE
Chose key VideoCode = 1, VideoTitle = 2 -> 1
Enter Video Code (5 digits) -> 02121
02121      DIRTY DANCING                                04

RUN OF INDEX-EG3.EXE USING VIDEOCODE
Chose key VideoCode = 1, VideoTitle = 2 -> 1
Enter Video Code (5 digits) -> 05051
05051      OVERBOARD                                    01

RUN OF INDEX-EG3.EXE USING VIDEOTITLE
Chose key VideoCode = 1, VideoTitle = 2 -> 2
Enter Video Title (40 chars) -> OVERBOARD
05051      OVERBOARD                                    01

RUN OF INDEX-EG3.EXE USING VIDEOTITLE
Chose key VideoCode = 1, VideoTitle = 2 -> 2
Enter Video Title (40 chars) -> DIRTY DANCING
02121      DIRTY DANCING                                04

RUN OF INDEX-EG3.EXE USING NON EXISTANT VIDEOCODE
Chose key VideoCode = 1, VideoTitle = 2 -> 1
Enter Video Code (5 digits) -> 44444
VIDEO STATUS :- 23
```

Select and Assign for Indexed Files

ENVIRONMENT DIVISION.

INPUT-OUTPUT SECTION.

FILE-CONTROL.

SELECT [OPTIONAL] FileName

ASSIGN TO FileSpec

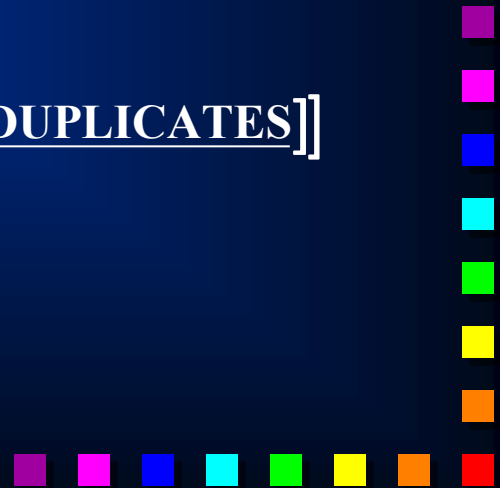
[ORGANIZATION IS] RELATIVE

[ACCESS MODE IS { SEQUENTIAL
 { RANDOM
 { DYNAMIC } }]

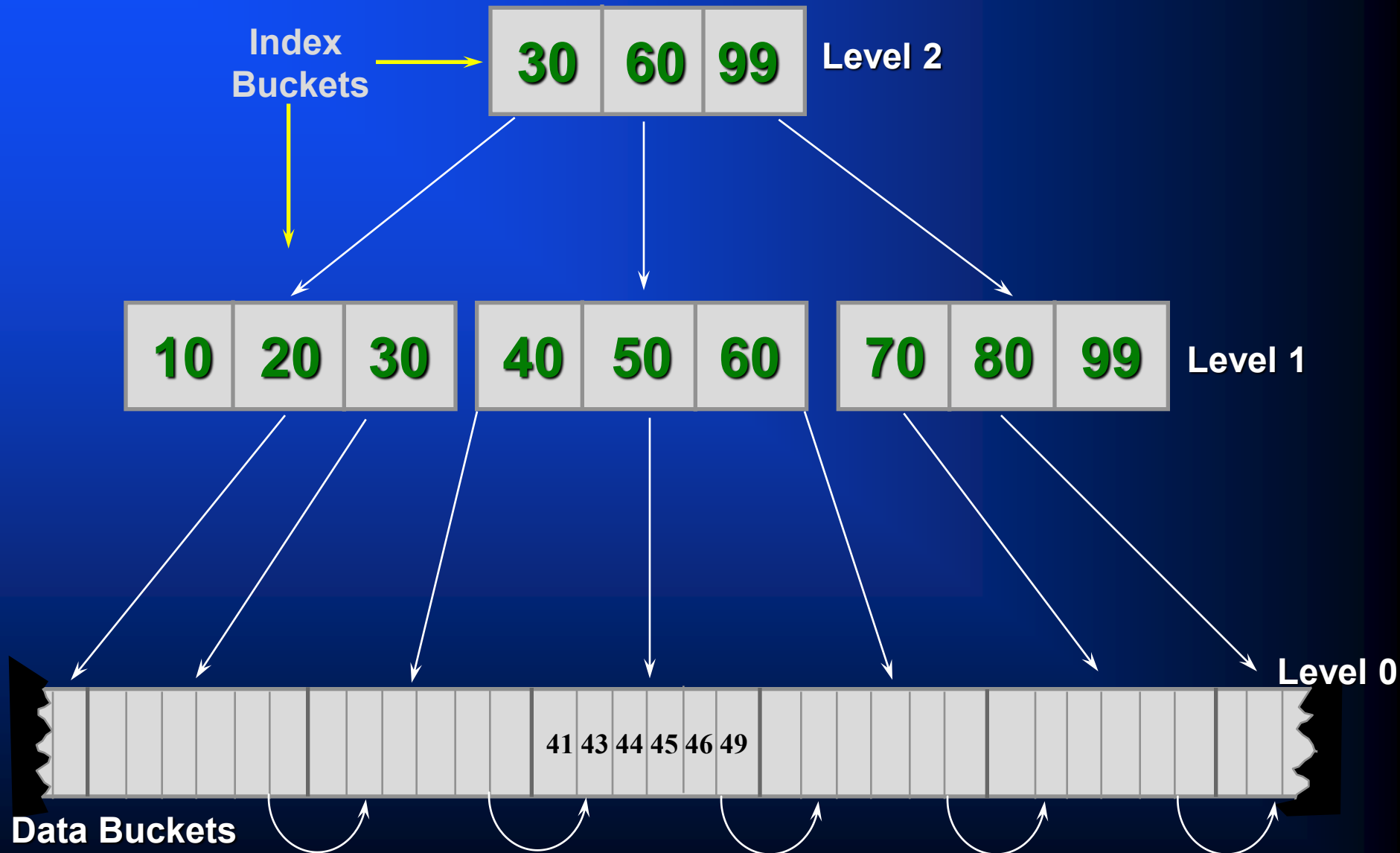
[RECORD KEY IS UniqueRecKey]

[ALTERNATE RECORD KEY IS AltKey [WITH DUPLICATES]]

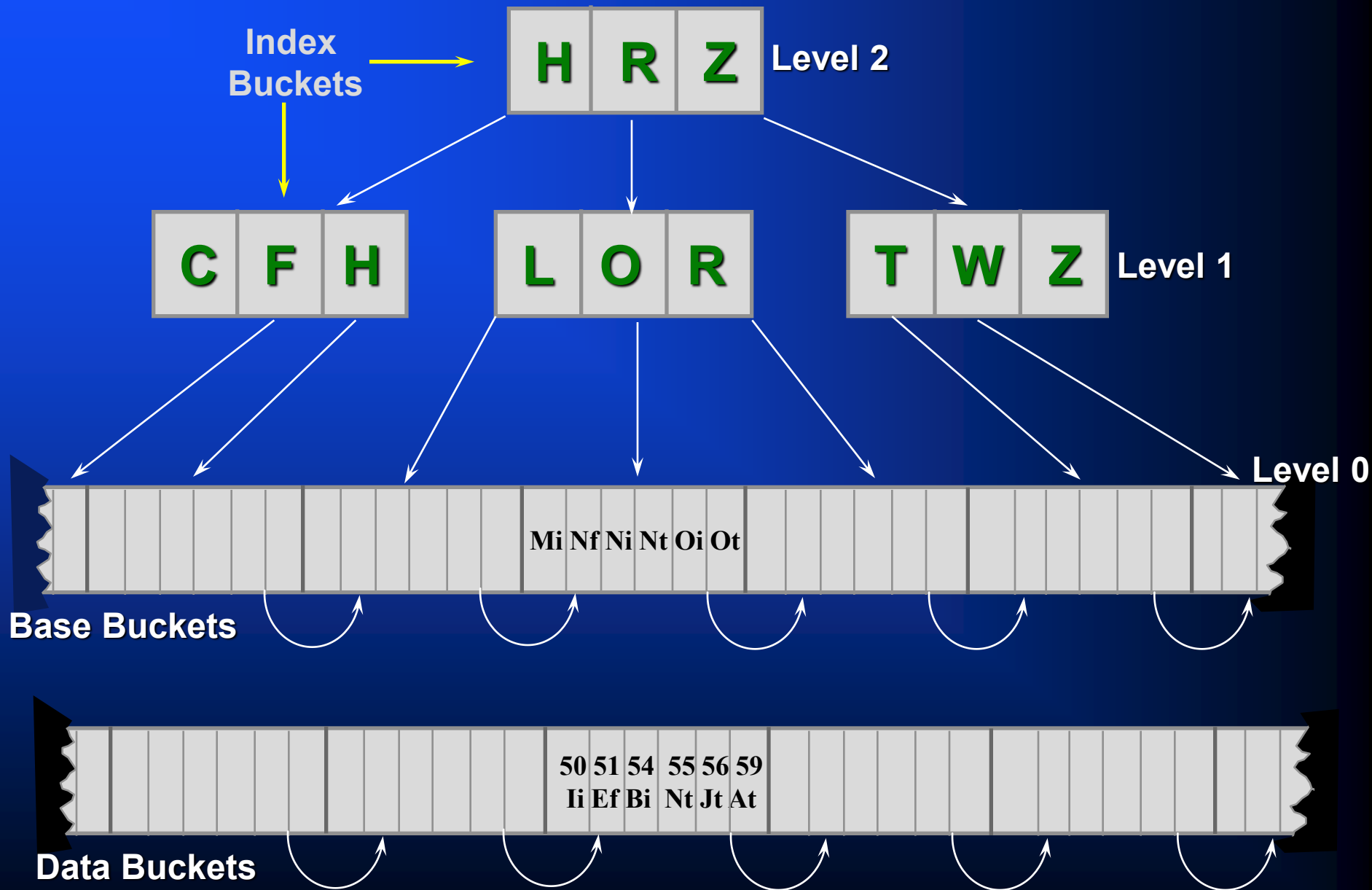
[FILE STATUS IS FileStatus]



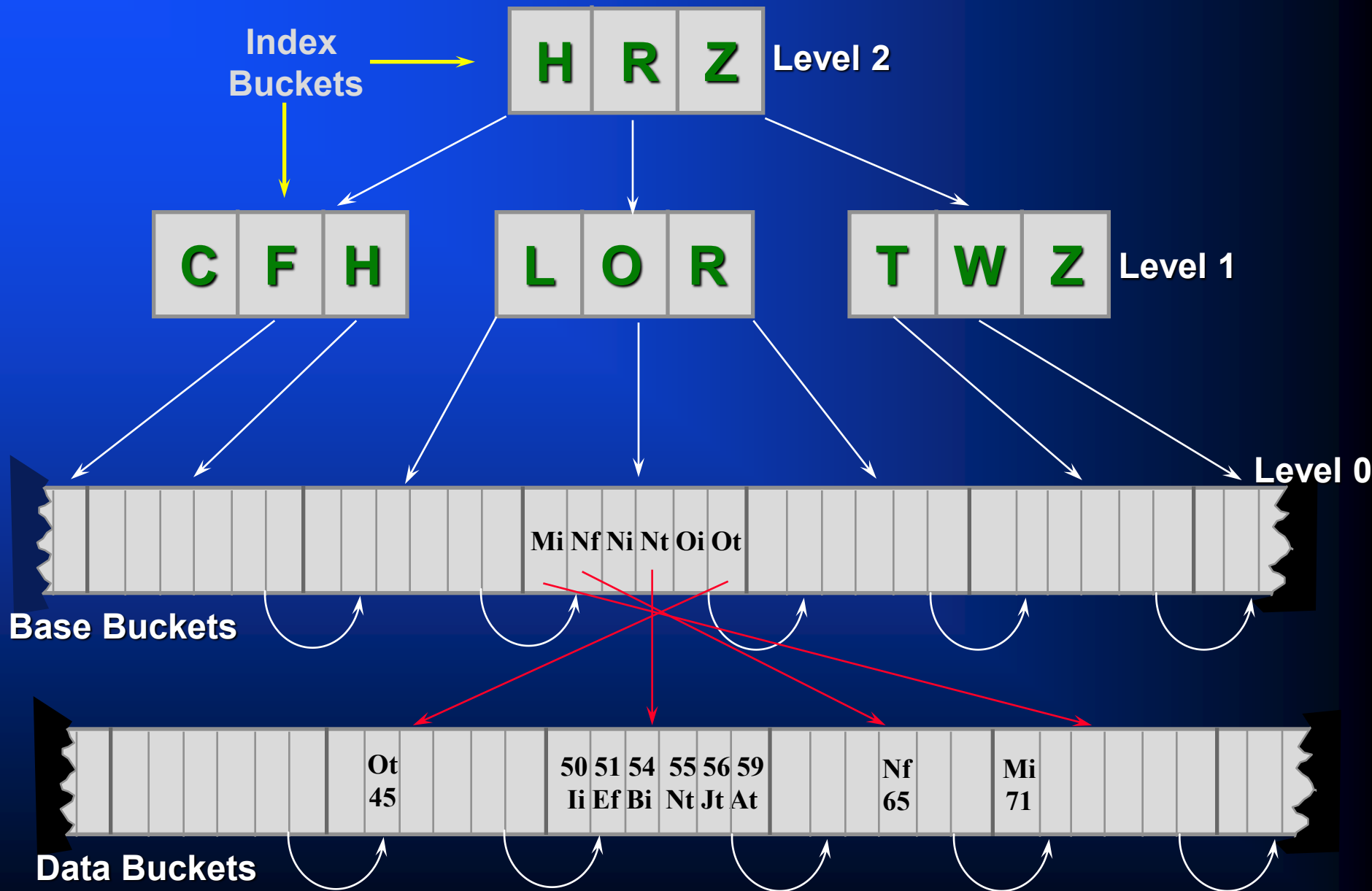
Indexed Files - Primary Key



Indexed Files - Alternate Key



Indexed Files - Alternate Key



FDs for Indexed Files

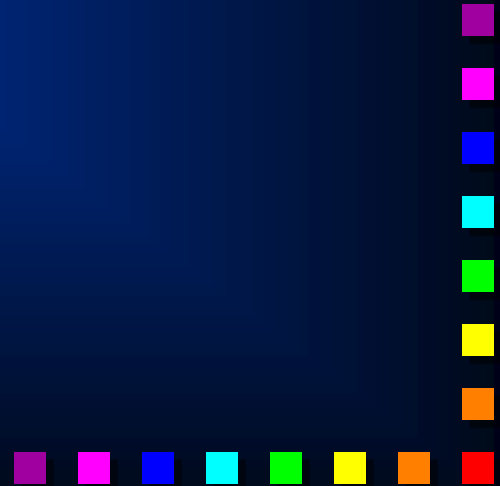
FD FileName

[IS EXTERNAL]

[IS GLOBAL]

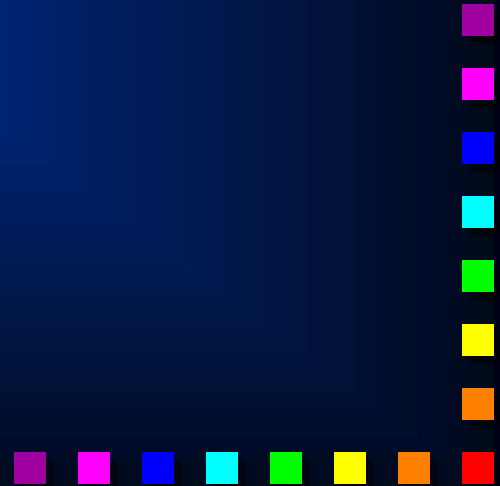
[BLOCK CONTAINS BlockSize {RECORDS
CHARACTERS}]

[VALUE OF ID IS FileSpec]



Indexed File Verbs - OPEN

OPEN { INPUT
OUTPUT
I-O
EXTEND } FileName ...



Indexed File Verbs - READ

READ FileName RECORD [INTO DestItem]
[KEY IS KeyName]
[INVALID KEY StatementBlock]
[END – READ]

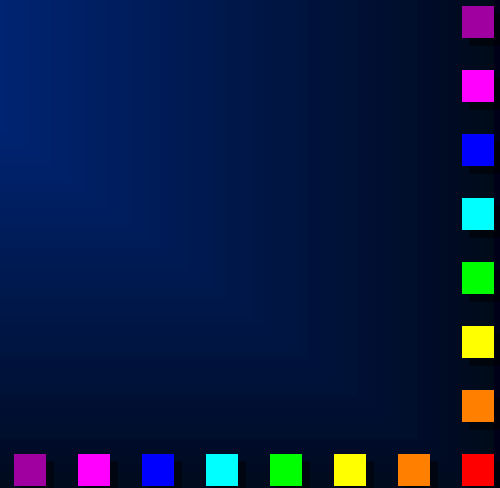
READ FileName NEXT RECORD [INTO DestItem]
[AT END StatementBlock]
[END – READ]



Indexed File Verbs - Write and Rewrite

WRITE RecName [FROM SourceItem]
[INVALID KEY StatementBlock]
[END – WRITE]

REWRITE RecName [FROM SourceItem]
[INVALID KEY StatementBlock]
[END – REWRITE]

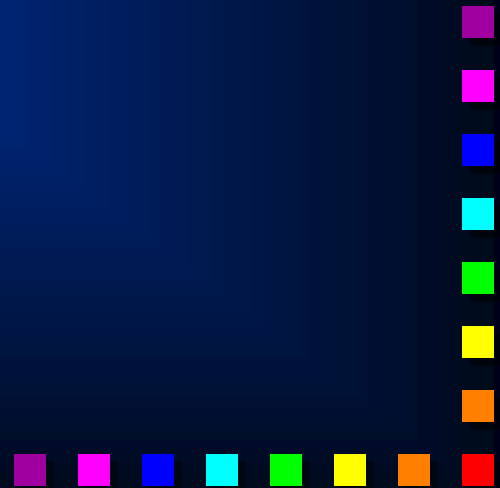


Indexed File Verbs - DELETE

DELETE FileName RECORD

[INVALID KEY StatementBlock]

END – DELETE



Indexed File Verbs - START

START FileName [KEY { IS EQUAL TO
IS =
IS GREATER THAN
IS >
IS NOT LESS THAN
IS NOT < } KeyName]
[INVALID KEY StatementBlock]

END – START

